

Affinoid G -Topologies, Measure Sheaves, and Tropical EM: A Research Proposal

Vinay K. Awasthi

vawasth1@jh.edu

Department of Applied Mathematics
Johns Hopkins University

September 20, 2026

Abstract

We propose a research program to establish a sheaf-theoretic foundation for non-Archimedean probability measures on affinoid G -topologies, and to construct an Expectation–Maximization (EM) framework on this structure. The central open problem is the existence of a Tate-type acyclicity theorem for measure sheaves valued in a non-Archimedean field. We identify which components already exist in the literature, which require modification, and which must be proved from scratch. The proposal is motivated by the cluster-prime Inside–Outside EM framework for hierarchical tensor grammars, where the resolution radius of a Berkovich skeleton plays the role of an amortization parameter.

1 Introduction and Motivation

1.1 Affinoid G -Topologies

Let K be a complete non-Archimedean field, and let A be an affinoid K -algebra with $X = \text{Max}(A)$ its maximal spectrum. The naive topology on X is totally disconnected and unsuitable for analytic geometry. Tate’s solution [1] was to introduce a *Grothendieck topology* (G -topology) in which:

- *Admissible open sets* are subsets $U \subset X$ admitting a set-theoretic cover by affinoid subdomains U_i such that for every K -affinoid algebra B and every morphism $f : A \rightarrow B$ with $f(\text{Max}(B)) \subset U$, the pullback cover $f^{-1}(U_i)$ has a finite subcover.
- *Admissible coverings* are covers by admissible opens satisfying an analogous finite-refinement condition under pullback.

These data satisfy axioms (T1)–(T4) of [2], making X a Grothendieck site.

The structure sheaf $\mathcal{O}_X$ is a sheaf for this G -topology. The foundational result is Tate’s acyclicity theorem:

Theorem 1 (Tate Acyclicity [4, Ch. 8], [?, Thm. 8.2.1]). *Let $X = \bigcup_i V_i$ be a finite affinoid covering, and let M be a finite Banach $\mathcal{O}_X$ -module. Then the Čech complex*

$$0 \rightarrow M \rightarrow \prod_i M_i \rightarrow \prod_{i,j} M_{ij} \rightarrow \cdots$$

is exact and admissible.

Theorem 1 applies only to *algebraic* sheaves ($\mathcal{O}_X$ -modules). It says nothing about measure sheaves.

1.2 Non-Archimedean Probability Measures

In the p -adic setting, σ -additivity is unavailable: every σ -additive $\mathbb{Q}_p$ -valued measure is discrete [6]. The correct substitute is *Monna–Springer boundedness*:

Definition 1 (Bounded additive measure [7]). *A measure $\mu : \mathcal{B} \rightarrow \mathbb{Q}_p$ on a ring of sets $\mathcal{B}$ is bounded if*

$$\|\mu\| := \sup_{A \in \mathcal{B}} |\mu(A)|_p < \infty.$$

A p -adic probability measure is a bounded additive measure with $\mu(\Omega) = 1$.

The Monna–Springer integral theory [7] provides the analytic foundation. Conditional expectation in this setting is a *ball*, not a point [8].

1.3 Measures as Sheaves on Grothendieck Sites

Recent work has begun to recast measure theory in sheaf-theoretic terms:

- Lehner [9] constructs the measurable locale of a Radon measure *directly from a Grothendieck topology on the poset of compact sets*, bypassing σ -algebras.
- Simpson [10] shows that random variables form an atomic sheaf on the category of standard Borel probability spaces.
- Grosu [11] equips a probability space $(\Omega, \mathcal{F}, P)$ with a Grothendieck topology τ_P induced by P , making it a filtered site.

No work combines these constructions with the affinoid G -topology.

1.4 The Cluster-Prime Application

The motivating application is the cluster-prime Inside–Outside EM of the companion paper. There, a rank-3 or rank-4 rule tensor $P_l(p \rightarrow x_1, \dots, x_m)$ is compressed to a small set of cluster primes, and the Inside recursion operates on cluster primes rather than individual symbols. The resolution radius r of the Berkovich skeleton is the amortization knob. The present proposal asks: *can this compression be made canonical via a measure sheaf on the affinoid G -topology?*

2 What Is Already Established

Component	Status	Reference
Affinoid G -topology axioms	Established	Chenevier [2]
Tate acyclicity (algebraic sheaves)	Established	BGR [4], Berkovich [3]
p -adic probability (bounded additive)	Established	Khrennikov [6]
Monna–Springer integration	Established	Monna–Springer [7]
Measure as sheaf (Radon case)	Established	Lehner [9]
Probability as atomic sheaf	Established	Simpson [10]
Probability as filtered site	Established	Grosu [11]

Table 1: Established components.

3 What Can Be Modified

3.1 Lehner's Construction over Affinoid Subdomains

Lehner [9] constructs measures from a Grothendieck topology on the poset of *compact subsets* of a Hausdorff space. The modification required is:

Replace the poset of compact subsets with the poset of *admissible affinoid subdomains* of X .

The question is whether the resulting locale carries a Monna–Springer measure in a way compatible with the admissible covering condition.

3.2 Simpson's Construction over Admissible Opens

Simpson [10] works over the category of standard Borel probability spaces. The modification required is:

Replace the base category with the category of admissible opens of the affinoid G -topology, with the atomic Grothendieck topology replaced by the G -topology.

The condition-independence characterization of the sheaf condition must be checked in the p -adic bounded-additive setting.

3.3 Grosu's Filtered Site over Affinoids

Grosu [11] equips a probability space with a Grothendieck topology induced by P . The modification required is:

Show that the topology induced by a Monna–Springer measure on an affinoid is compatible with the admissible coverings of the G -topology.

4 What Must Be Proved

4.1 Theorem A: Existence of the Measure Sheaf

Conjecture 1 (Measure sheaf). *Let $X = \mathrm{Sp}(A)$ be a K -affinoid space, and let μ be a Monna–Springer bounded additive measure on X . Then there exists a sheaf $\mathcal{F}_\mu$ on the affinoid G -topology such that for every admissible open $U \subset X$, $\mathcal{F}_\mu(U)$ is a localization of μ to U .*

Difficulty. Monna–Springer measures are defined on the *compact-open* ring, while admissible opens need not be compact-open. A compatibility argument is required.

4.2 Theorem B: Acyclicity

Conjecture 2 (Acyclicity). *For every admissible affinoid covering $X = \bigcup_i U_i$ and every measure sheaf $\mathcal{F}_\mu$, the Čech complex*

$$0 \rightarrow \mathcal{F}_\mu(X) \rightarrow \prod_i \mathcal{F}_\mu(U_i) \rightarrow \prod_{i,j} \mathcal{F}_\mu(U_i \cap U_j) \rightarrow \cdots$$

is exact, i.e. $\check{H}^j = 0$ for $j > 0$.

This is the central obstacle. Tate acyclicity (Theorem 1) does not apply, because the values of $\mathcal{F}_\mu$ lie in a non-Archimedean field, not in an affinoid algebra.

4.3 Theorem C: EM Projections

Conjecture 3 (EM projections). *On the measure sheaf $\mathcal{F}_\mu$, there exist m -projections (E -step) and e -projections (M -step) that converge monotonically to a local optimum.*

Difficulty. Standard information geometry relies on the convexity of KL divergence. In the p -adic setting, KL divergence is unavailable; the candidate replacement is the tropical divergence

$$D_{\text{trop}}(\mu \parallel \nu) = \sum_B \mu(B) \cdot v_p\left(\frac{\mu(B)}{\nu(B)}\right).$$

The tropical Pythagorean identity must be verified.

5 The Three Core Theorems

We isolate the three theorems that form the backbone of the program. If all three are proved, the affinoid measure-sheaf EM framework is essentially complete. They are listed in dependency order: Theorem 1 is foundational, Theorem 2 is the central obstacle, Theorem 3 is the algorithmic payoff.

5.1 Theorem 1: Existence of the Measure Sheaf

Theorem 2 (Measure sheaf on the affinoid G -topology). *Let K be a complete non-Archimedean field, $X = \text{Sp}(A)$ an affinoid K -space, and μ a Monna–Springer bounded additive measure on the compact-open ring $\Delta(X)$ of X . Then there exists a sheaf $\mathcal{F}_\mu$ on the affinoid G -topology such that:*

- (i) *For every admissible affinoid subdomain $U \subset X$, $\mathcal{F}_\mu(U)$ is a K -module canonically associated to $\mu|_U$;*
- (ii) *The restriction maps $\mathcal{F}_\mu(U) \rightarrow \mathcal{F}_\mu(V)$ for admissible $V \subset U$ are compatible with the localization of μ ;*
- (iii) *$\mathcal{F}_\mu$ is a sheaf for admissible coverings: for every admissible covering $U = \bigcup_i U_i$, the sequence*

$$0 \rightarrow \mathcal{F}_\mu(U) \rightarrow \prod_i \mathcal{F}_\mu(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}_\mu(U_i \cap U_j)$$

is exact in the category of K -modules.

Why first. Without the sheaf, there is nothing to do EM on. This is the foundational object.

Key difficulty. Monna–Springer measures live on the *compact-open* ring $\Delta(X)$, while admissible opens need not be compact-open. The construction must localize μ to arbitrary admissible opens and verify the sheaf condition.

Sub-problems.

- (a) Prove that every admissible affinoid subdomain U carries an induced Monna–Springer measure $\mu|_U$.
- (b) Prove that restriction maps are well-defined and compatible with the p -adic topology.
- (c) Verify the sheaf condition on admissible coverings (which must satisfy conditions (T1)–(T4), not arbitrary covers).

Closest existing work. Lehner’s [9] construction of measurable locales from Grothendieck topologies on compact-set posets. Specialization to the affinoid case is required.

5.2 Theorem 2: Tate-Type Acyclicity for Measure Sheaves

Theorem 3 (Acyclicity of the measure sheaf). *Let $X = \mathrm{Sp}(A)$ be an affinoid K -space, $\mathcal{F}_\mu$ the measure sheaf of Theorem 2, and $\mathfrak{U} = \{U_i\}_{i \in I}$ a finite admissible affinoid covering of X . Then the Čech complex*

$$0 \rightarrow \mathcal{F}_\mu(X) \rightarrow \prod_i \mathcal{F}_\mu(U_i) \rightarrow \prod_{i,j} \mathcal{F}_\mu(U_i \cap U_j) \rightarrow \prod_{i,j,k} \mathcal{F}_\mu(U_i \cap U_j \cap U_k) \rightarrow \cdots$$

satisfies

$$\check{H}^j(\mathfrak{U}, \mathcal{F}_\mu) = 0 \quad \text{for all } j > 0.$$

Moreover, the complex is admissible: the image of each differential is closed and the cohomology is Hausdorff in the p -adic topology.

Why second. This is the central obstacle identified in the proposal.

Key difficulty. Tate’s acyclicity theorem (Theorem 1) applies to *algebraic* sheaves ($\mathcal{O}_X$ -modules) whose values lie in affinoid algebras. The measure sheaf $\mathcal{F}_\mu$ takes values in a non-Archimedean field K , not an affinoid algebra, so Tate’s proof does not apply directly.

Sub-problems.

- (a) Show that every admissible covering can be refined to a covering by *rational subdomains* $U(f_1, \dots, f_n/g) = \{|f_i| \leq |g|\}$.
- (b) Prove acyclicity for rational coverings using Kiehl descent or a Monna–Springer convergence argument.
- (c) Verify that p -adic boundedness of μ controls convergence of the Čech complex at each degree.
- (d) Handle non-Archimedean convergence: Čech differentials involve infinite sums over I that must converge in the p -adic topology.

Closest existing work. Tate’s original proof for $\mathcal{O}_X$ -modules [4, 3], and Kedlaya’s [12] reduction to Laurent coverings. Neither handles measures.

Strategic note. Start with the simplest case: $X = \{|z| \leq 1\}$, μ the Monna–Springer measure, and $\mathfrak{U}$ the Laurent covering $\{|z| \leq 1\} = \{|z| \leq 1\} \cup \{|z| = 1\}$. This is the base case of Tate’s proof. If acyclicity holds here, you have the seed of the general argument.

5.3 Theorem 3: Tropical EM Projections and Convergence

Theorem 4 (Tropical EM). *Let $\mathcal{F}_\mu$ be the measure sheaf of Theorem 2, and suppose Theorem 3 holds so that $\mathcal{F}_\mu$ is acyclic. Define the tropical divergence between two measures μ, ν on the affinoid G -topology by*

$$(\mu \| \nu) = \sum_{B \in} \mu(B) \cdot v_p \left(\frac{\mu(B)}{\nu(B)} \right),$$

where the sum is over the Berkovich skeleton at resolution r , and v_p is the p -adic valuation. Then:

- (i) **(Tropical Pythagorean identity)** *For any measure ν and any m -projection μ^* of ν onto the model manifold,*

$$(\nu \| \mu^*) + (\mu^* \| \mu) = (\nu \| \mu)$$

for all $\mu \in$.

- (ii) (**Monotonic convergence**) The iterated E-step (m-projection) and M-step (e-projection) produce a sequence $\mu^{(k)}$ such that the tropical energy

$$^{(k)} := \sum_B \mu^{(k)}(B) \cdot v_p(\mu^{(k)}(B))$$

is monotone non-increasing: $^{(k+1)} \leq ^{(k)}$.

- (iii) (**Convergence**) The sequence $\{\mu^{(k)}\}$ converges in the p -adic weak topology to a fixed point of the EM iteration.

Why third. This is the algorithmic payoff. Without it, Theorems 2 and 3 are structural results with no computational content.

Key difficulty. Standard information geometry relies on the convexity of KL divergence. In the p -adic setting, KL divergence is unavailable; the tropical divergence replaces it. The Pythagorean identity must be re-proved in the tropical setting.

Sub-problems.

- (a) Prove that the tropical divergence is well-defined on the Berkovich skeleton and takes values in $\mathbb{R} \cup \{+\infty\}$.
- (b) Prove the tropical Pythagorean identity by reducing to the piecewise-linear structure of the skeleton.
- (c) Prove monotone convergence of the tropical energy, replacing classical log-likelihood monotonicity.
- (d) Handle the resolution parameter r : as $r \rightarrow 0$, the skeleton refines. Show that the EM fixed point is stable under refinement, or identify the convergence rate.

Closest existing work. Amari's information geometry for the classical case; the tropical Pythagorean identity appears in fragmentary form in tropical geometry, but not for measures on G -topologies. This is the most novel part of the program.

Strategic note. The tropical Pythagorean identity is the load-bearing lemma. Prove it first for the two-vertex skeleton, then induct on the number of vertices.

5.4 Summary of the Three Theorems

Theorem	Role	Difficulty	What breaks if it fails
2 Measure sheaf exists	Foundational	Medium	No object to work with
3 Acyclicity	Central obstacle	High	EM has no Čech-theoretic meaning
4 Tropical EM projections	Algorithmic payoff	Medium–High	No convergence guarantee

Table 2: The three core theorems, in dependency order.

Prove them in order. Theorem 3 is the hardest, but it depends on Theorem 2. Theorem 4 depends on Theorem 3. Attempting Theorem 3 before Theorem 2 is solid will waste time on a moving target.

6 Technical Path

6.1 Step 1: From Compact-Open to Admissible Opens

Monna–Springer measures live on the compact-open ring $\Delta(X)$. Affinoid admissible opens need not be compact-open. Prove that every admissible open can be approximated by compact-open sets in a way compatible with boundedness.

6.2 Step 2: Sheaf Definition

Define

$$\mathcal{F}_\mu(U) := \varprojlim_{K \subset U, K \text{ compact-open}} \mu(K),$$

with the limit taken in the p -adic topology. Verify:

- (i) The presheaf is separated.
- (ii) Gluing holds over admissible coverings.
- (iii) Boundedness is preserved.

6.3 Step 3: Acyclicity

Mimic Tate’s strategy: first prove for rational coverings, then reduce general admissible coverings to Laurent coverings. The obstruction is that Tate’s reduction uses the affinoid algebra structure, which is absent for measure sheaves. Find an analogous reduction.

6.4 Step 4: EM Projections

If Conjectures 1 and 2 hold, construct the E-step and M-step as in the companion paper, but with the tropical divergence replacing KL. Verify monotonic convergence.

7 Risk Assessment

Risk	Likelihood	Mitigation
Acyclicity fails	Medium	Weaken to finite covers
Tropical Pythagorean fails	High	Abandon projection, use coordinate descent
Admissible/compact-open incompatibility	Medium	Restrict to paracompact affinoids
EM does not converge	Low	Standard EM convergence on compact spaces

Table 3: Risk assessment.

8 Expected Contributions

If successful, the program would:

- (i) Establish the first sheaf-theoretic foundation for non-Archimedean measures on affinoid G -topologies.

- (ii) Prove a Tate-type acyclicity theorem for measure sheaves.
- (iii) Construct a tropical information-geometric EM framework.
- (iv) Provide a bridge between p -adic analytic geometry and probability theory.

References

- [1] J. Tate, *Rigid analytic spaces*, Invent. Math. **12** (1971), 257–289. (Notes written 1962.)
- [2] G. Chenevier, *Lecture notes on rigid analytic geometry*, §1.12–1.15.
- [3] V. Berkovich, *Spectral Theory and Analytic Geometry over Non-Archimedean Fields*, AMS Math. Surveys and Monographs **33**, 1990.
- [4] S. Bosch, U. Güntzer, R. Remmert, *Non-Archimedean Analysis*, Springer, 1984. (Chapters 8–9.)
- [5] J. Werner, *Coherent Sheaves on Non-Archimedean Analytic Spaces*, Thesis.
- [6] A. Khrennikov, *p -adic valued probability measures*, Indag. Math. **7**(3) (1996), 311–330.
- [7] A. Monna and T. Springer, *Intégration non-archimédienne I, II*, Indag. Math. **25** (1963), 634–653, 654–661.
- [8] S. Evans, *Expectation, conditional expectation and martingales in local fields*, arXiv:math/0606609, 2007.
- [9] G. Lehner, *Measure theory via locales*, arXiv:2510.08826, 2025.
- [10] A. Simpson, *Probability sheaves*, arXiv:2405.11073, 2024.
- [11] O. Grosu, *Algebraic stochastic calculus*, arXiv:1407.6784, 2014.
- [12] K. Kedlaya, *MIT 18.727: Topics in Algebraic Geometry — Rigid Analytic Geometry*, Lecture notes, esp. “More on G-topologies.”
- [13] M. Masdeu, *Lecture notes on rigid analytic geometry*, Tate acyclicity proof.
- [14] A. Ducros, *Les espaces de Berkovich sont modérés*, arXiv:1010.2235.